

DOES THE COMBINED USE OF CLASS AND SEQUENCE DIAGRAMS IMPROVE THE SOURCE CODE COMPREHENSION? RESULTS FROM A CONTROLLED EXPERIMENT

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Motivation

The wide diffusion of the UML

due to the number of visual notations it offers to write system blueprints, including conceptual and concrete items

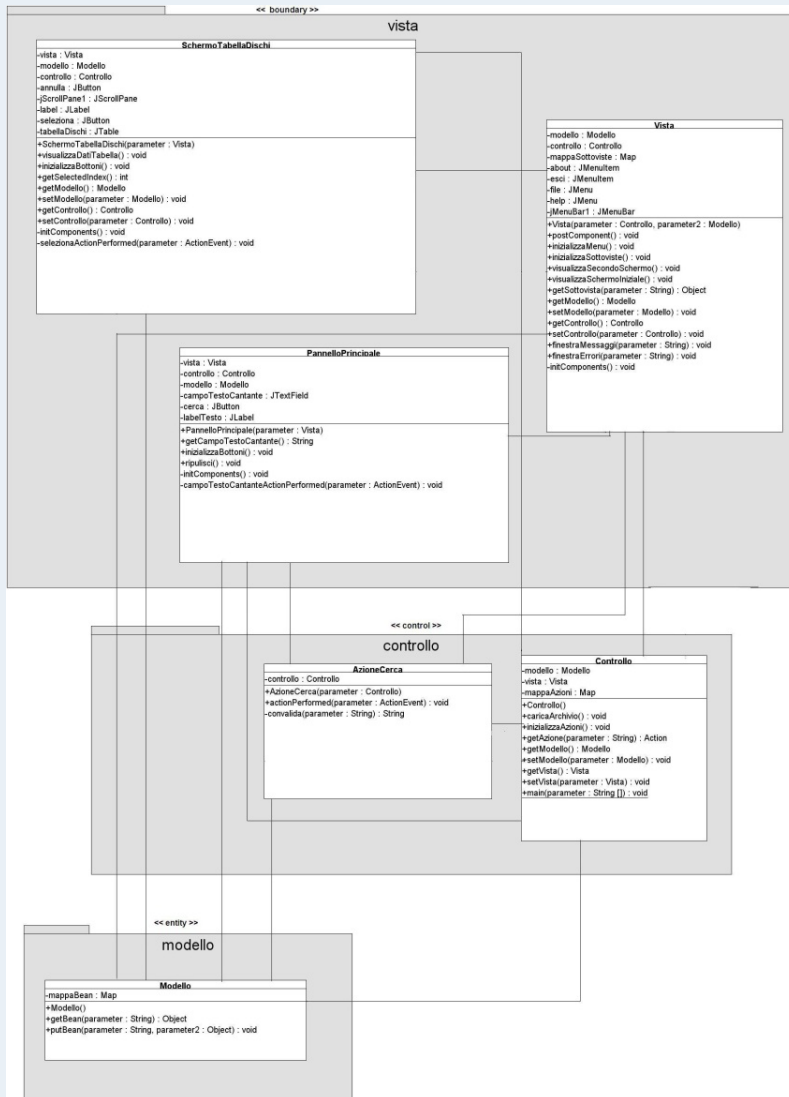
Benefit deriving from the use of UML in all the phase of the software life cycle

Only few studies to assess the support of the UML models (produced in the design phase) in the comprehension of source code

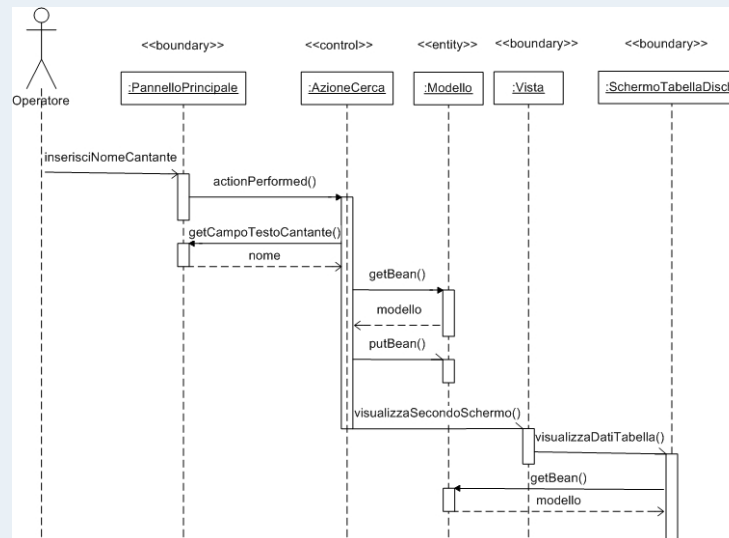
E. Arisholm, L. C. Briand, S. E. Hove, and Y. Labiche. The impact of UML documentation on software maintenance: An experimental evaluation. *IEEE Transactions on Software Engineering*, 32:365-381, 2006.

W. J. Dzidek, E. Arisholm, and L. C. Briand. A realistic empirical evaluation of the costs and benefits of UML in software maintenance. *IEEE Transactions on Software Engineering*, 34:407-432, May 2008

Motivation (2)



Do UML class and sequence diagrams produced in the design phase improve documentation and maintenance of software systems?



Definition

Analyze the use of UML class and sequence diagrams produced in the design phase

for the purpose of evaluating them

with respect to their support in the comprehension of source code

from the point of view of the researcher, **in the context of** students in Computer Science

from the point of view of the project manager, **in the context of** junior programmers

Context selection

16 Master students from University of Salerno

Basic software engineering knowledge (requirement engineering, high and low level design of OO software systems based on UML, software development, and software maintenance.

Not graded on the experimental tasks

Two systems: *Music Shop* (MS) and *Theater Ticket Reservation* (TTR)

Never underwent maintenance operations

Experimental objects: two chunk of MS (S1 of 463LOCs) and TTR (S2 of 378LOCs)

MS class diagram: 6 classes, 27 attributes, and 45 methods

MS sequence diagram: 1 actor, 5 objects, and 11 messages

TTR class diagram: 5 classes, 22 attributes, and 36 methods

TTR sequence diagram: 1 actor, 5 objects, and 9 messages

Hypotheses Formulation and Variables

Hn0. The presence of UML class and sequence diagrams does not significantly improve the comprehension of source code.

One-tailed (we expected a positive effect of UML models)

Independent variable: **Method**

two possible values: NO_Mo (source code alone) and Mo (source code added with UML diagrams)

Dependent variable: **Comprehension**

the number of correct answers provided by the participants

Experimental task

We asked to fill in a comprehension questionnaire composed of 15 multiple choice questions

Each admitting 3 possible answers with only one correct

Questions to assess several aspects related to the comprehension of the source code, e.g.:

In case the user selects a ticket for an already booked chair, what is the method invoked and what is the message shown to the user?

What is/are the class/es defined to manage new exceptional conditions about “BuyTicket” and the loading of information about the tickets?

Experiment design and co-factors

	Group A	Group B	Group C	Group D
Run 1	S1, Mo	S1, NO_Mo	S2, Mo	S2, NO_Mo
Run 2	S2, NO_Mo	S2, Mo	S1, NO_Mo	S1, Mo

Within-participant counterbalanced experimental design

Co-factors:

System. Differences in the experimental objects (e.g., complexity and domain) could affect **Comprehension**

Order of Method. The order in which the participants performed the laboratory runs may bias the results

Experimental procedure

Pilot: to evaluate possible issues related to the experimental material

Experiment:

- i) Introductory lesson
- ii) Execution of the two comprehension tasks
 - No time limit
- iii) Post-experiment

Analysis procedure

R for statistical analysis:

Descriptive statistics to summarize the results

Mann-Whitney test to test the null hypothesis , and the effect of System and Order of Method (5% of committing Type-I-error)

Statistical Power to analyze the probability that a statistical test will reject a null hypothesis when it is actually false

Boxplots to graphically show the answers to post-survey questionnaire

Results

System	Observation	Mo			NO_Mo		
		Mean	Median	Std. Dev.	Mean	Median	Std. Dev.
All	32	13.06	14	1.53	11.75	12	1.29
S1	16	13.12	14	1.25	11.62	11.5	1.41
S2	16	13	13.5	1.85	11.88	12	1.25

Participants achieved significant better Comprehension values when using Mo

High statistical power ($t=0.91$)

No significant effect of:

System and Order of Method

Further analysis on **Completion time**

the combined use of UML class and sequence diagrams does not significantly affect the time the participants spent to comprehend source code

Post Experiment Questionnaire

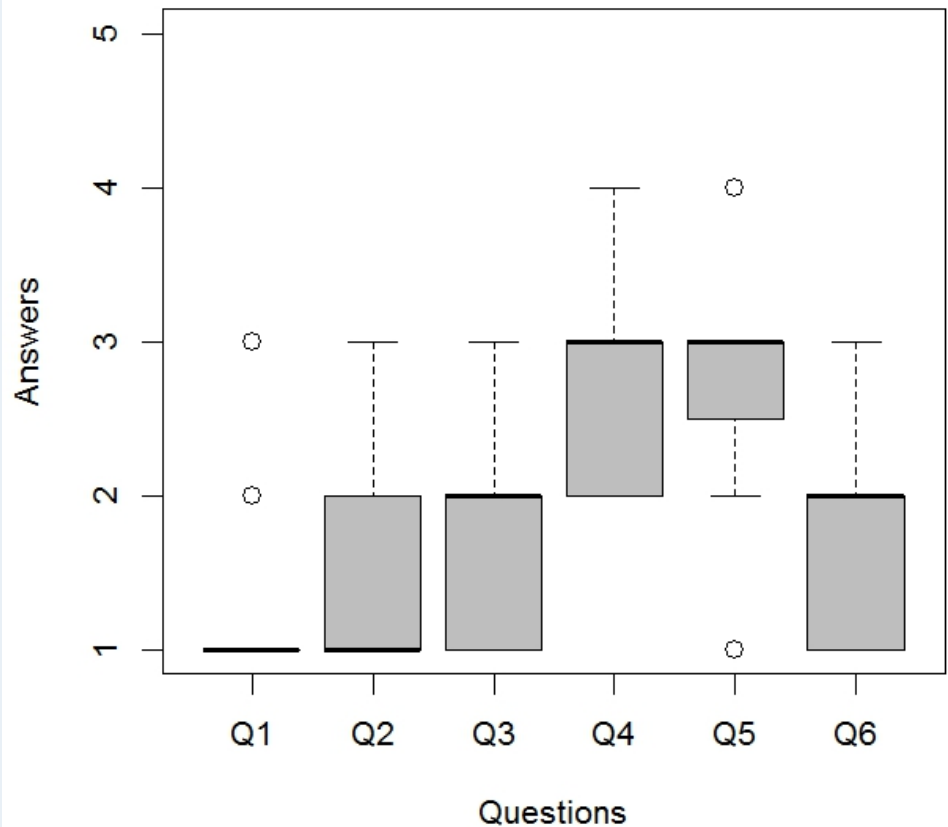
Time appropriate

Task objectives clear

Tasks clear

Neutral judgment about the complexity of exp. objects

The UML diagrams are useful



Threats to validity

Internal validity

the participants had similar experience with the UML and software modeling

the order in which the participants performed the tasks did not significantly affect source code comprehension

The exchange of information among participants was prevented

Conclusion validity

null hypothesis rejected with p-value less than 0.01 and 0.91 as the statistical power value

Number of participants

Threats to validity (2)

Construct validity

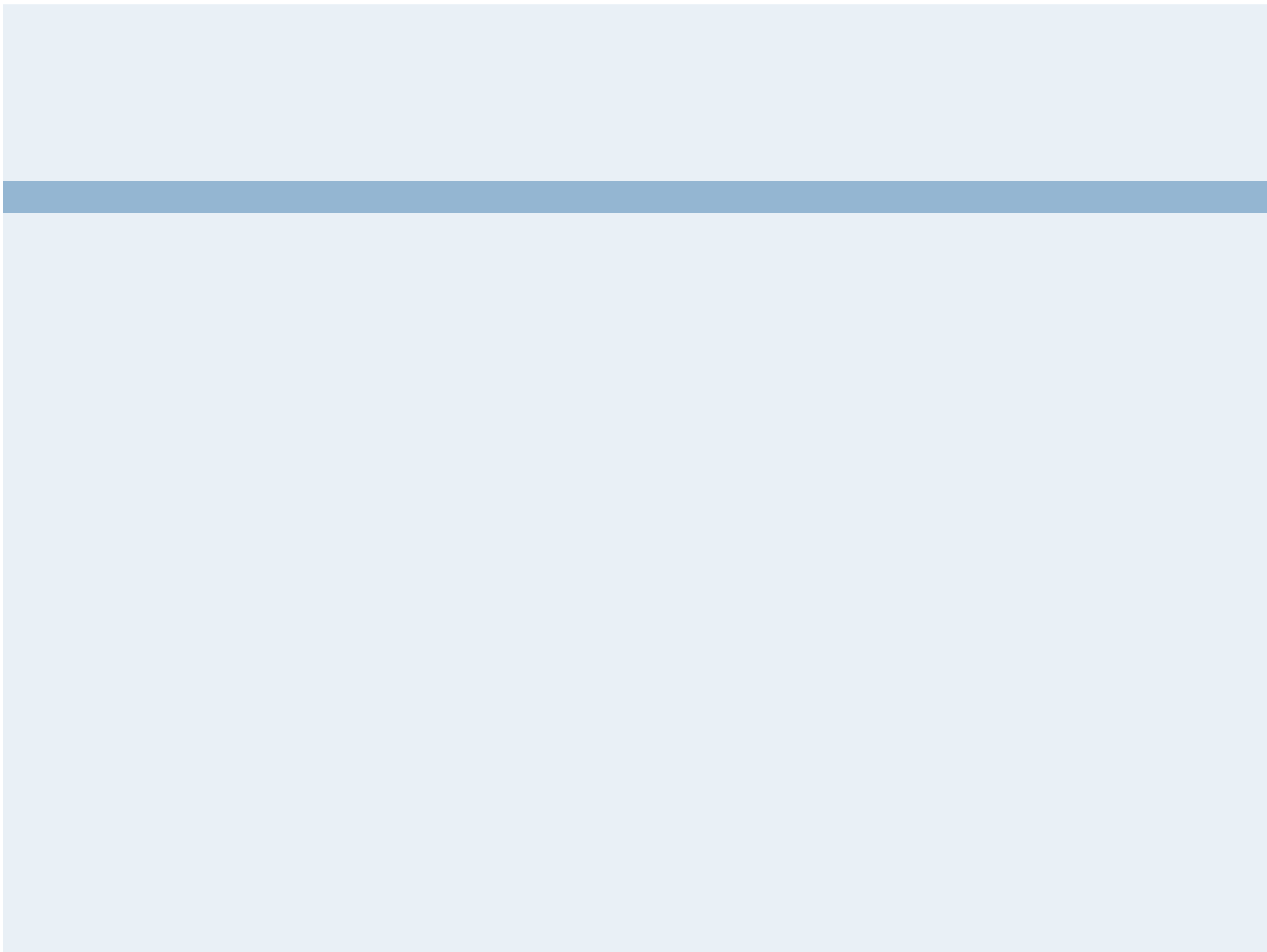
Evaluation apprehension: students were not evaluated on the performances

The measure for dependent variable well known and used

External validity

The use of students

Size and complexity of the objects



Order of Method

Diff(NO Mo) the differences for the Comprehension values achieved by the participants, who performed the tasks with NO_Mo first and then with Mo;

Diff(Mo) the differences for the Comprehension values achieved by the participants who performed the tasks with Mo first and then with NO_Mo.

Mann-Whitney U exact test to verify

Had: $\text{Diff}(\text{NO_Mo}) > \text{Diff}(\text{Mo})$

Bonferroni

To investigate the effect of System, we applied the Bonferroni correction. In particular, we used two tests to analyze whether the effect of System is statistically significant. Then, the p-values have to be less than $5\%/2 = 2.5\%$